KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY
(AN AUTONOMOUS INSTITUTION)
Accredited by NAAC, Approved by AICTE, Affiliated to JNTUH, Hyderabad

Department of Computer Science & Engineering (AI&ML)
A.Y: 2026-27

Intelligent Medical Image Analysis with Automated Clinical Report Generation

Abstract:

Medical image analysis plays a vital role in the early detection and diagnosis of diseases. This project proposes an AI-based clinical decision support system capable of analyzing multiple medical image modalities, including X-rays, CT scans, MRI scans, skin lesion images, and retinal images. The system employs a vision model to identify the anatomical region, detect diseases or abnormalities, localize affected areas, estimate severity, and provide confidence scores for predictions. Patient information such as age, gender, symptoms, and medical history is integrated with the image analysis results to provide more meaningful clinical insights. A language model is then used to generate a structured medical report summarizing the diagnostic findings and recommending the appropriate medical specialist for further consultation. To improve interpretability and user experience, the system may also incorporate RetrievalAugmented Generation (RAG) or API-based integration to retrieve relevant medical knowledge and generate concise, context-aware report summaries. By combining automated image analysis, explainable AI techniques, intelligent report generation, and specialist recommendation into a unified framework, the proposed system aims to support healthcare professionals in improving diagnostic accuracy, reducing reporting time, and enhancing clinical decision-making.

Keywords:
Medical Image Analysis, Artificial Intelligence, Vision Model, Disease Detection, Abnormality Localization, Severity Estimation, Explainable AI, Language Model, Retrieval-Augmented Generation (RAG), Clinical Decision Support, Automated Medical Report Generation

Team No: 13

Domain of the project: AI&ML + Medical

Team Members:

S.No	Roll No	Student Name	Student Sign
1	**23BD1A6636**	**M.Hrushikesh**	
2	**23BD1A6639**	**N.Ramana**	
3	**23BD1A663B**	**N.Vinesh**	
4	**23BD1A6627**	**B.Mallesh**	

Faculty Supervisor **Project Coordinator**